Dual-Laser PDH Driver

Two-channel CW 1550 nm DFB laser + TEC driver for differential ring-resonator thermometry • Project summary, rev 3 (RF derivative lock)

What the instrument does

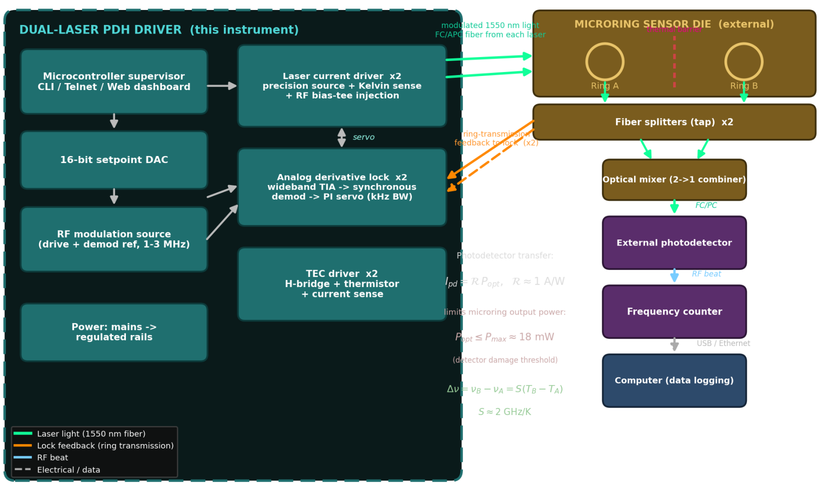
The driver locks two independent DFB lasers, one to each of a pair of 1550 nm ring resonators that sit on a **single silicon die separated by a thermal barrier**. Each laser is held on its ring's transmission slope by a closed feedback loop. The measured quantity is the **optical beat between the two locked lasers**, which is a direct, ratiometric readout of the *local temperature difference* across the barrier:

$$\Delta\nu = \nu_B - \nu_A = S\cdot(T_B - T_A), \quad \text{with } S = d\nu/dT \approx 2\ \text{GHz/K}$$

Because the result is a material-defined ratio, the scale factor and all slow common-mode drifts cancel, so absolute accuracy and long-term reproducibility are not required. The figure of merit is **within-window stability over ~60 s**, projected at the single-to-low-tens of nanokelvin level. Each laser channel comprises a precision current source with low-side Kelvin current sensing, a thermo-electric cooler loop with thermistor and current sensing, and a **1–3 MHz analog derivative lock** (RF current modulation → wideband transimpedance amplifier → synchronous demodulator → analog PI servo). An embedded microcontroller supervises acquisition, calibration, and telemetry.

Optical interfacing. Each laser delivers light over an **FC/APC** (angled) single-mode fiber connector, chosen to suppress back-reflection into the DFB source; the beat photodetector terminates in an **FC/PC** fiber connector. The photodetector converts ring output power to current as I_{pd} **= □·P_{opt}** (responsivity □ ≈ 1 A/W), and its handling limit caps the usable power out of the microrings: $P_{opt} \le P_{max}$ **≈ 18 mW** (detector damage threshold). This sets the optical-power budget for the ring outputs feeding the beat path.

Measurement setup



Each laser is current-locked to its microring via the ring-transmission feedback path. A fiber splitter and an optical mixer sit **inline with the microring outputs**: the splitters tap each ring output for the lock feedback, and the through arms are combined in the 2→1 mixer onto the external photodetector. The resulting RF beat is read by a frequency counter whose output goes to an **external computer** for logging — not back into the instrument. The two channels run at slightly different modulation frequencies so modulation crosstalk stays out of the differential beat.

Estimated build cost

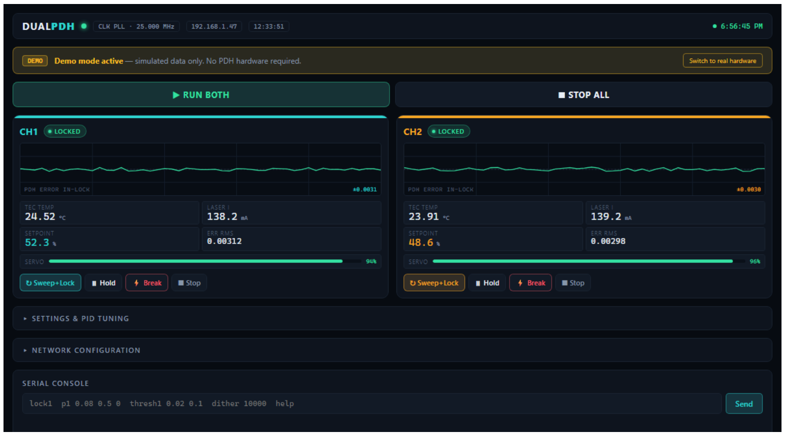
Item	Detail	Qty	Est. cost (USD)
Board electronics (DigiKey-orderable)	Op-amps, DACs, DDS, TIAs, mixers, regulators, passives	1 set	$552
DFB laser modules	1550 nm CW butterfly laser, FC/APC fiber output	2	$3,600
Fiber photodiodes	InGaAs fiber photodiode, FC/PC connector	2	$240
Bare parts subtotal			**$4,392**
PCBs	Fabrication + assembly allowance		$1,000
Mechanical fixture / enclosure	Optical mounting + chassis allowance		$1,000
ESTIMATED TOTAL			**$6,392**

Small-quantity estimates from the KiCad-derived BOM (15 hierarchical sheets). The two DFB laser modules dominate cost; board electronics are a small fraction of the total.

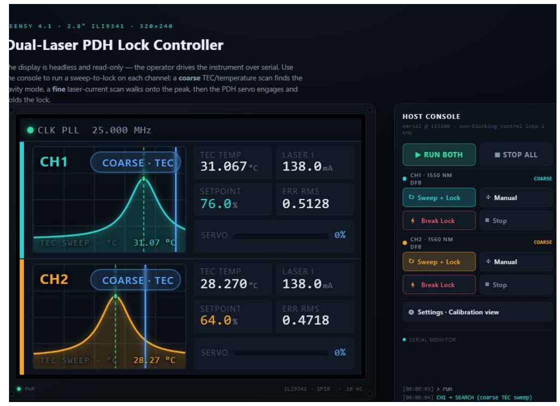
Control software & operator interface

The instrument is driven by embedded microcontroller firmware that acts as a **supervisor** over the analog lock hardware. Per channel it programs the modulation and phase-reference oscillators, sweeps the DC setpoint to acquire the cavity (a coarse TEC/temperature scan finds the mode, a fine laser-current scan walks onto the peak), auto-calibrates the demodulation phase, hands the loop to the analog PI servo, and continuously monitors the post-mix error signal and lock quality. A built-in simulation (demo) mode reproduces the full acquire-and-lock behaviour with no hardware attached. The same control surface is exposed three ways:

USB-serial / Telnet CLI — text command console (also embedded in the web page and the front panel). **HTTP dashboard** — live browser UI on port 80. **On-board TFT** — 320×240 read-only status display.



Web dashboard: per-channel error trace, TEC temperature, laser current, setpoint, servo health, lock state, and an embedded serial console.



On-board TFT: coarse/fine scan view and live lock status for both channels.

Basic interface functions

Command	Function
lock1 / lock2	Start the sweep-and-lock acquisition sequence on a channel
stop1 / stop2	Return a channel to idle (outputs to safe midscale)
hold1 / hold2	Freeze the servo and outputs at their current values
break1 / break2	Deliberately break the lock and re-enter the relock search
omega1 / omega2 <Hz>	Set the RF modulation frequency per channel (100 kHz - 3.5 MHz)
phase1 / phase2 <deg>	Set the demodulation reference phase
cal1 / cal2	Auto-calibrate the demod phase for maximum error-signal slope
thresh1 / thresh2 <l> <a>	Set capture / loss thresholds for lock detection
s / status	Print full telemetry (state, error RMS, temp, currents, gains)
d	Raw ADC dump of all monitor channels
demo on / demo off	Toggle hardware-simulation vs. real-hardware mode (saved to EEPROM)
netset / netinfo / netreset	Configure and apply DHCP/static IP, gateway, and mDNS hostname

Graphical controls on the web dashboard and front panel map onto the same commands: **Run Both / Stop All**, and per-channel **Sweep+Lock**, **Hold**, **Break**, and **Stop**, plus collapsible panels for PID/threshold tuning and network configuration. Lock state is colour-coded (search → acquire → locked / relock) on every interface.

Uncertainty budget (this design)

Differential ring-resonator thermometer • ratiometric, within-60 s window • RF analog-derivative lock

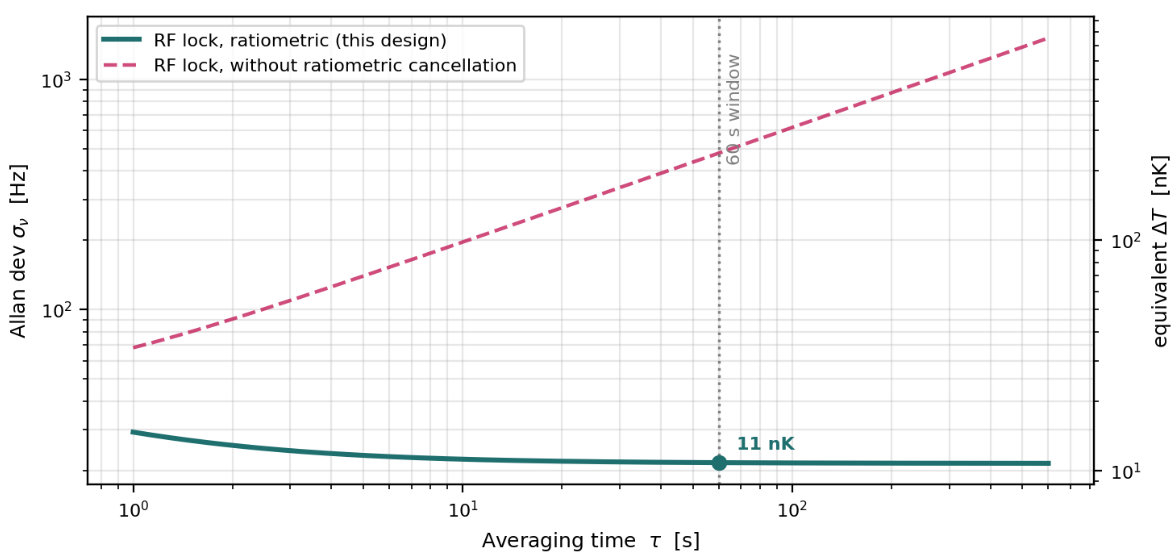
The measurement is a **ratio defined by the die material** ($\Delta\nu = S \cdot \Delta T$, $S \approx 2$ GHz/K), acquired over a ~60 s window. Because it is ratiometric, the scale factor and every slow common-mode term cancel, so the figure of merit is **within-window stability** — the Allan deviation at $\tau \approx 60$ s, expressed as an equivalent differential temperature. The budget below is for **this design only**: the 1–3 MHz analog-derivative (RF) lock. All entries are 1σ per channel; terms flagged *cancels* drop out of the ratio and do not enter the result.

Within-window budget at τ = 60 s — RF analog-derivative lock

Contributor	In the ratio	σν @ 60 s	Equiv. ΔT
Detection / discrimination noise	kept	2.6 Hz	1.3 nK
Residual laser FM (above loop BW)	kept	6.0 Hz	3.0 nK
RIN → lock-point pulling (in-band)	kept	5.0 Hz	2.5 nK
In-band RAM (flicker)	kept	20.0 Hz	10.0 nK
Carrier-modulation smear	kept	~0	~0
Timebase short-term	kept	0.01 Hz	~0
RAM slow drift	*cancels*	*(387 Hz)*	*(194 nK)*
Thermal common-mode leakage	*cancels*	*(232 Hz)*	*(116 nK)*
Strain / mount creep	*cancels*	*(155 Hz)*	*(77 nK)*
Combined (ratiometric) @ 60 s	**—**	**21.6 Hz**	**10.8 nK**

Kept terms add in quadrature to the combined result; the cancelled terms (shown in parentheses) are large but are common-mode and removed by the ratio. **In-band RAM dominates the surviving budget.** If the residual amplitude modulation is slower than the 60 s window it too cancels and the floor drops to **~4 nK**; with ~20 Hz of in-band flicker RAM the floor is **~11 nK**. Measuring the RAM spectrum early is the key open item that fixes where in this 4–11 nK band the instrument lands.

Beat-frequency Allan deviation



Solid: this design (RF lock) with ratiometric cancellation — the curve stays flat across the 60 s window at the low-nanokelvin floor. Dashed: the same lock *without* ratiometric cancellation, where slow common-mode terms re-enter as a random-walk upturn (hundreds of nK by 60 s). The ratiometric, same-die measurement is what makes the low-nK figure reachable without an absolute frequency reference.

Operative result: with ratiometric cancellation the within-window spec is ~4–11 nK at $\tau \approx 60$ s, set by in-band RAM. Reproduced from the project differential-thermometer uncertainty budget (1σ, $S \approx 2$ GHz/K).